

ND-120 FPGA Debug Reference Card

Basys3 (xc7a35tcbg236-1) • 100 MHz • 115200 Baud • Branch: redo-idb

LED Layout (left LD15 to right LD0)

TERM LD15	CC3 LD14	CC2 LD13	CC1 LD12	CC0 LD11	— LD10	— LD9	— LD8	LCS LD7	MCLK LD6	BEAT LD5	UART LD4	RST LD3	RUN LD2	GRN LD1	RED LD0
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Switches (SW0=rightmost on Basys3)

Switch	Signal	UP (1)	DOWN (0)
SW0 (right)	sys_rst_n	Reset released, CPU runs	CPU held in reset
SW1	btn2	7-seg = MAC address	7-seg = MIC address

LED Details

ND-120 CPU • Norsk Data 1988 • FPGA Implementation • 2026-03-30

LED	Signal	ON	OFF
LD15	~TERM_n	Cycle termination	-
LD14-11	~CC3..0_n	Cycle counter (cycles)	-
LD7	~LCS_n	Microcode loaded	Loading
LD6	MCLK	Memory clock	No MCLK
LD5	clkTick[26]	Blinks 1.5Hz	No clock!
LD4	~uartTx	TX active	Silent
LD3	sys_rst_n	Reset released	In reset
LD2	~s_run	CPU running	OPCOM
LD1	cpu_led[1]	GREEN	-
LD0	cpu_led[0]	RED (err)	OK

7-Segment Display (4 hex digits)

SW1	Shows	Range
DOWN	MIC address (CSA_12_0, microcode)	0000 - 1FFF
UP	MAC address (LA_23_10, memory)	0000 - 3FFF

Changing values = CPU executing • Stuck 0000 = CPU not running

UART

115200 Baud	8N1 Format	None Flow	A17 TX pin	A18 RX pin
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Boot Sequence

- 1 Program bitstream
- 2 LD5 blinks (clock running)
- 3 Slide SW0 UP (release reset)
- 4 LD3 ON (reset released)
- 5 LD15-11 cycling (state machine)
- 6 LD7 ON after ~6ms (microcode loaded)
- 7 LD2 ON (CPU running)
- 8 7-seg shows changing hex
- 9 Terminal shows # prompt

Vivado ILA Debug Signals (mark_debug)

Signal	W	Description
s_debug_csa[12:0]	13	MIC address (microcode fetch)
s_debug_la_23_10[13:0]	14	MAC address upper (LA 23:10)
s_debug_ca_9_0[9:0]	10	MAC address lower (CA 9:0)
s_debug_cc_term[4:0]	5	{TERM,CC3,CC2,CC1,CC0}_n
s_debug_mclk	1	Memory clock
s_debug_lcs_n	1	LCS_n (0=loading, 1=loaded)
s_run	1	CPU run (0=running)
s_debug_uartTx	1	UART TX line
s_debug_uartRx	1	UART RX line
s_debug_cpu_led[6:0]	7	CPU status LEDs

Deep Hierarchy (add via Set Up Debug)

Path	W	Description
CPU_BOARD/.../s_FIDB0_15_0	16	CGA data bus output
CPU_BOARD/.../s_FIDBI_15_0	16	CGA data bus input
CPU_BOARD/CPU/CS/s_csbits	64	TOPCSB microcode word
CPU_BOARD/.../s_idb_erb_out	16	Register file to IDB
CPU_BOARD/.../CHIP_32H/txState	3	UART TX state machine
CPU_BOARD/.../CHIP_32H/rxState	3	UART RX state machine
CPU_BOARD/CYC/s_cc0_n	1	Cycle counter bit 0

Pin Map

Port	Pin	Port	Pin	Port	Pin
sysclk	W5	led[5]	U15	led[12]	P3
btn1/SW0	V17	led[6]	U14	led[13]	N3
btn2/SW1	V16	led[7]	V14	led[14]	P1
uartTx	A17	led[8]	V13	led[15]	L1
uartRx	A18	led[9]	V3	seg[0-6]	W7..U7
led[0]	U16	led[10]	W3	an[0-3]	U2..R2
led[1]	E19	led[11]	U3		
led[2]	U19				
led[3]	V19				